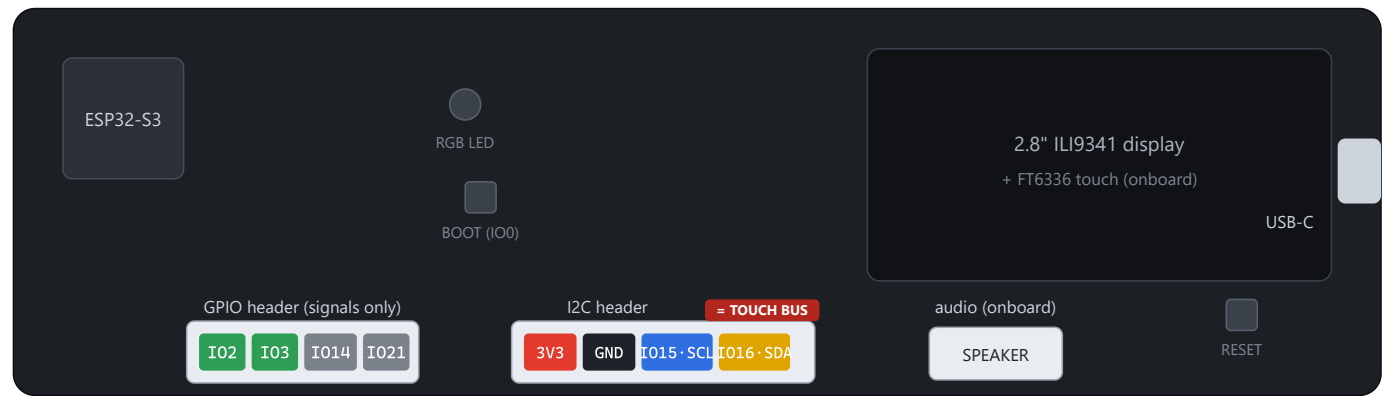


MindBox — Wiring Guide

LCDWIKI ES3C28P / ES3N28P · 2.8" ESP32-S3 touch board · how every component connects

To just make it run: plug in the **USB-C cable**. The display, capacitive touch, RGB LED and audio amp are already on the board — nothing else has to be wired. Everything below is only for *adding sensors* (e.g. your microphone).

1 · The board at a glance



Power for sensors: take 3V3 + GND from the I2C header.
Green pins (I02/I03) are the only analog-capable (ADC1) pins on the headers. **Grey** pins (I014/I021) are digital-only.

Top view, matching the build photo: two 4-pin headers + the SPEAKER connector along the bottom edge.

2 · Connector pinouts

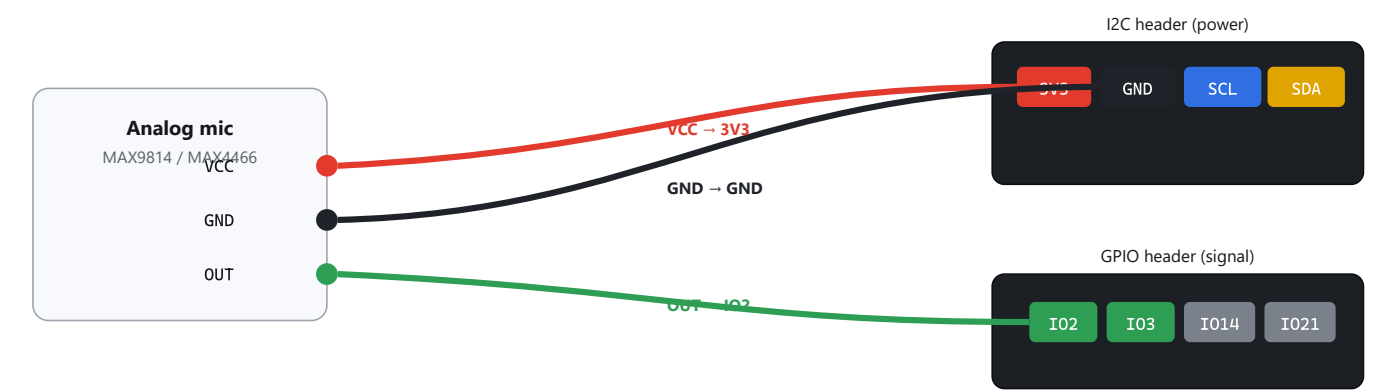
Connector	Pins (left → right)	Use
GPIO header (left 4-pin)	I02 I03 I014 I021	General I/O. No power pins here. Analog only on I02/I03.
I2C header (middle 4-pin) TOUCH BUS	3V3 GND I015 SCL I016 SDA	Use 3V3 + GND to power sensors. SCL/SDA are the capacitive-touch bus (FT6336 @ 0x38) — not a free I2C port.
SPEAKER (2-pin)	+ -	Onboard audio amp output. Firmware doesn't use audio yet.
USB-C	5V + USB	Power + flashing/serial. The only cable needed to run the unit.

3 · Already on the board — do not wire

Component	Internal connection
2.8" ILI9341 display	SPI — SCK IO12 · MOSI IO11 · MISO IO13 · DC IO46 · CS IO10 · backlight IO45
FT6336 capacitive touch	I2C @ 0x38 — SDA IO16 · SCL IO15 · INT IO17 · RST IO18
Audio amp + SPEAKER	Onboard I2S amp (IO4–IO8 region) → SPEAKER connector
RGB LED · USB-C · RESET · BOOT	Onboard (BOOT = IO0: tap = theme, hold = recalibrate touch)

4 • Wiring your microphone (analog module)

An analog mic (MAX9814 / MAX4466) outputs a voltage, so it needs an ADC1 pin — and IO2/IO3 are the only ADC1 pins on the headers.



Power (3V3/GND) comes from the I2C header; the signal wire (OUT) goes to IO2 on the GPIO header.

Mic pin	Board pin	Header
VCC	3V3	I2C header · pin 1
GND	GND	I2C header · pin 2
OUT	IO2	GPIO header · pin 1

Then in firmware

```
// src/config.h
#define PIN_MIC 2 // OUT on IO2 (ADC1)
#define HAS_MIC 1 // turn the mic on
```

Reflash, then in Serial Monitor (115200): press **c** to calibrate, **m** to watch the level live. This powers the “noise high” alert + the Noise setting.

5 • Optional add-ons (for later)

Component	Wire to	config.h	Notes
Light sensor (KY-018, analog)	OUT→IO3, VCC→3V3, GND→GND	PIN_LIGHT_ADC 3, HAS_LIGHT 1	ADC1 pin; can’t share IO2 with the mic.
DHT11 temp/humidity	DATA→IO21, VCC→3V3, GND→GND	PIN_DHT11 21, HAS_TEMP 1	Add a 4.7k–10k pull-up from DATA to 3V3.
VL53L1X presence (I2C)	SDA→IO16, SCL→IO15, VCC→3V3, GND→GND	HAS_PRESENCE 1	Shares the touch bus (addr 0x29 ≠ 0x38). Needs firmware to read it on the touch I2C port — ask first.
Haptic motor	gate→IO21 via transistor + flyback diode; motor on its own 5V	PIN_HAPTIC 21	Never drive a motor straight from a GPIO.

Two analog sensors at once → one on **IO2**, the other on **IO3** (the only two ADC1 pins on the header).

6 • Rules & traps

DO power sensors from **3V3**.

DO put analog sensors on **IO2 / IO3** (ADC1).

DO set the pin + flip `HAS_*` in `config.h`, reflash, verify with serial `m/d/c`.

DON'T use 5V for a sensor — it can clip/damage the ADC pin.

DON'T put analog on IO14/IO21 (not ADC1 — wrong readings).

DON'T hang a generic I2C device on the "I2C" header — it's the touch bus. And the radio is **2.4 GHz only**.

7 · Build / flash settings (Arduino IDE)

Board **ESP32S3 Dev Module** · USB CDC On Boot = **Enabled** · Flash **16MB** · Partition **Huge APP (3MB No OTA/1MB SPIFFS)** · **PSRAM** = **OPI PSRAM** · Serial **115200**. Wi-Fi is set on the device under **Device** → **Wi-Fi Setup** (scan + on-screen keyboard), or over serial with `w`.

MindBox firmware · pin facts authoritative in `src/config.h` · see also `docs/HARDWARE.md`, `docs/WIRING.md`.